Nishant Saravanan

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OBJECTIVE

- Electrical and Electronics Engineering student seeking Summer 2027 hardware internships and research opportunities, applying hands-on experience in circuit design, PCB layout, embedded systems, sensor integration, and power electronics

EDUCATION

University of Georgia, College of Engineering – Athens, GA — Aug 2026 – May 2029

Bachelor of Science in Electrical and Electronics Engineering — GPA: 4.0

Relevant Coursework: Intro to ECSE, Physics: Electricity and Magnetism, Differential Calculus, Logic Design

PROJECT EXPERIENCE

Soil Health Monitoring Probe, Innovation Academy – Alpharetta, GA — Dec 2024 – May 2026

- Designed the probe's sensor circuit and Autodesk Fusion CAD enclosure around an ESP8266 microcontroller, integrating a DHT11 (temperature, humidity), NPK sensor (nitrogen, phosphorus, potassium), and pH sensor to monitor soil health
- Programmed the ESP8266 in C++ (Arduino) to sample sensor data and transmit it over Wi-Fi to a live dashboard, integrating an AI detection model to flag soil disease risk and report soil health results to the farm owners in real time
- Field-tested across 2 acres for 12 weeks, iterating on the design to cut power draw with ESP8266 deep-sleep mode, improve enclosure durability, and report additional soil metrics, while identifying further refinements for future design versions

Automated Anesthesia System Demonstration, Innovation Academy – Alpharetta, GA — Aug 2025 – May 2026

- Designed a prototype automated anesthesia system that read real-time heart rate, blood pressure, ECG, and body temperature data, combining live vitals with prior patient information to calculate and display dosage levels
- Built a demonstration unit that visually output the calculated dosages to show how the system adjusts to changing patient vitals, and presented the working project at our high school's engineering showcase in front of local employers

PROFESSIONAL EXPERIENCE

Information Technology Intern, VDart Inc. – Alpharetta, GA — Aug 2025 – Dec 2025

- Used n8n workflow automation and AI agents to research and evaluate 15+ potential market spaces for VDart to explore
- Built a Moodle Cloud-based web application enabling 60+ prospective employees to earn a professional job certification
- Developed an engagement app with surveys, learning modules, and interactive content connecting young adults in India with business, education, and career resources while gathering demographic data to guide VDart's program outreach

IT High School Intern, Fulton County Schools IT Department – Roswell, GA — May 2025 – Aug 2025

- Reimaged and configured 2,000+ computers across the Fulton County School District, wiping outdated data and installing standard software configurations to prepare all devices for students, teachers, and staff for the upcoming school year
- Resolved ServiceDesk tickets and managed district inventory with Windows Autopilot, enrolling and assigning devices

CAMPUS INVOLVEMENT & ASSOCIATIONS

LEARNSat Hardware Engineer, Small Satellite Research Laboratory – Athens, GA — Sep 2026 – Present

- Contributing to the electronics of LEARNSat, SSRL's low-cost, 3D-printed CubeSat kit for K-12 schools, supporting power distribution, sensor payload integration, PCB design, and hardware testing for student-assembled satellites
- Writing hardware documentation and assembly guides for the LEARNSat Ed curriculum used by high school CubeSat teams

First-Year Council, Institute of Electrical and Electronics Engineers – Athens, GA — Aug 2026 – Present

- Developing a Phantom Load Killer with a team of 4 EE undergraduates, a smart inline outlet that detects standby (phantom) power draw from idle devices and automatically cuts power to eliminate wasted energy
- Attending 12+ events per semester, including company engagement sessions, technical workshops, and social events

TECHNICAL PROFICIENCIES

Hardware: Altium Designer, Autodesk Fusion, Circuit Design, PCB Design, Soldering, Sensors, Multimeter, Power Supplies
Embedded: ESP8266, ESP32, Arduino, Raspberry Pi, C++ Firmware, Sensor Integration, Wi-Fi IoT, Low-Power Design
Software: Python, C++, Java, MATLAB, JavaScript (React), Multisim, LTspice, n8n, Moodle Cloud, Windows Autopilot